

The Rate Without the Extent: a Machine-Checked Uniform Spectral Modulus for the Decoupled $\mathbb{Z}_2$ Slice and a pre-registered judge that failed at the coupled one

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Abstract

Every rate in this lane so far has been a fixed-extent rate. The gap paper proved strict spectral separation at each extent and said plainly that it was not uniform [2]; the modulus paper gave that separation a number, $\text{specRatio}(L)$, and reported measurements saying the number tends to 1 outside the disordered region [3]. A geometric bound whose rate tends to 1 is empty in the volume limit, so nothing in the lane survived $L \rightarrow \infty$, and the word *clustering* was never used.

Proved. For the *decoupled* kernel — the transfer kernel at constant source weight — the modulus is $\text{specRatio} = \tanh \beta$ at *every* extent, with L nowhere in it. Both directions: an operator bound by induction on the extent, and attainment by the single-site observable the extent paper already built [1]. Composing with the modulus paper’s endpoint, the normalised Gibbs two-point function obeys $|\mathbb{E}[A(X_0)A(X_N)]| \leq C_A (\tanh \beta)^N$ past one threshold serving every observable — a bound whose *rate* contains no L , and therefore the first statement in this lane that survives the volume limit.

Why the proof is not the spectral decomposition. The decoupled kernel is a product over sites, so its spectrum is a product; that route needs the spectrum of a Kronecker power, which the library does not carry. It is not needed. The modulus paper proved that specGap is the *greatest* norm ratio on the fluctuation sector, so bounding it above is an operator inequality and nothing else, and the operator inequality falls to induction: the even part of an observable keeps its mean zero and inherits the rate, the odd part keeps nothing and gets only Schur’s test, and the two recombine *exactly*, because $\tanh \beta \cdot Z = D$ with Z the row sum and D the odd eigenvalue of a single bond.

Not proved, and a judge that failed. The *coupled* kernel is untouched. Before any of this was written we pre-registered two falsifiable predictions. The first — that the decoupled rate is exactly $\tanh \beta$ at every extent — passed to 10^{-16} , and authorised the work above. The second — that the coupled uniformity boundary is the Onsager curve — failed on one of eight pre-registered cells, and it stays failed: that claim is therefore reported as **not established**, not softened. At constant source weight the spatial slices are independent, so what is proved here is a statement about a product measure; that is exactly why it is reachable, and it is said here rather than left to be noticed. Reflection positivity is untouched, and nothing in this paper is a claim about $\text{SU}(N)$, the continuum limit, or the Yang–Mills mass gap [6].

1 Ten papers of fixed-extent rates

The lane’s rates have all been local in the extent. That is not a stylistic complaint: a rate $r(L)$ with $r(L) \rightarrow 1$ gives, at fixed N , a bound that becomes vacuous as the system grows, and the

infinite-volume statement one actually wants — exponential decay of correlations with a fixed rate — is not implied by any amount of fixed- L information.

So the honest question is whether *some* kernel in this family has a rate that does not move with L . This paper answers it for the decoupled kernel, and the answer is exact.

2 The construction

Let $Z = e^\beta + e^{-\beta}$ and $D = e^\beta - e^{-\beta}$, and let K be the decoupled kernel on $\{0, 1\}^L$: a product over sites of the single-bond matrix. Its row sums are all Z^L , so the constant observable is its Perron vector with eigenvalue $\lambda = Z^L$.

Lemma 2.1 (Schur’s test). *A symmetric nonnegative kernel all of whose row sums equal R satisfies $\|Ku\| \leq R\|u\|$.*

`norm_act_le_of_rowSum`. One Cauchy–Schwarz per row, then summing. This is the only bound the odd part of the induction is given, and it is all it needs.

Theorem 2.2 (the uniform operator bound). *For every extent L and every observable u with $\sum_\sigma u(\sigma) = 0$,*

$$\|Ku\| \leq \tanh \beta \cdot Z^L \cdot \|u\|.$$

`spatialKernel_fluct_bound`, from the squared form `spatialKernel_fluct_sq`, which is what the induction runs on — a square root in the inductive hypothesis would be a square root at every step.

3 Why the constant does not degrade

Split an observable at the first site, $u = (u_0, u_1)$, and set $p = u_0 + u_1$, $m = u_0 - u_1$. Then p still has mean zero, so the inductive hypothesis applies to it with rate $\tanh \beta$; m has no orthogonality left and gets only Lemma 2.1, with rate 1. Acting with the kernel and collecting,

$$\|Ku\|^2 = \frac{1}{2}(Z^2\|Kp\|^2 + D^2\|Km\|^2), \quad \|u\|^2 = \frac{1}{2}(\|p\|^2 + \|m\|^2).$$

Feeding in the two bounds gives $\frac{1}{2}Z^2(\tanh \beta Z^L)^2\|p\|^2 + \frac{1}{2}D^2(Z^L)^2\|m\|^2$, and the two coefficients are *equal*, because

$$\tanh \beta \cdot Z = D.$$

`tanh_mul_z2Norm`. That single identity is the whole reason the constant survives: the rate given up on the odd part is exactly the odd eigenvalue of one bond, at every step. Lose it and the constant would pick up a factor per site.

4 Sharpness

Theorem 2.2 is an inequality. It is attained.

Theorem 4.1 (the rate is exactly $\tanh \beta$). *At every extent with at least one site, $\text{specGap} = \tanh \beta \cdot Z^L$, hence $\text{specRatio} = \tanh \beta$.*

`spatialKernel_specGap_eq`. The upper bound is Theorem 2.2 together with the fact that specGap is the *greatest* norm ratio on the fluctuation sector [3] — which is the only place that paper is used, and it is used as a bound, never as a spectral decomposition. The lower bound is the single-site sign observable of the extent paper [1]: it has mean zero (`sum_siteSign`), it is nonzero, and its eigenvalue $D Z^{L-1}$ realises the bound exactly.

5 The payoff

Theorem 5.1 (a rate that survives the volume limit). *At constant source weight there is an N_0 such that for every observable A whose dressing has mean zero there is a $C_A > 0$ with*

$$|\mathbb{E}[A(X_0)A(X_N)]| \leq C_A (\tanh \beta)^N, \quad N \geq N_0,$$

and $\tanh \beta$ does not depend on the extent.

`gibbs_clustering_uniform_rate`, composing Theorem 4.1 with the modulus paper’s endpoint [3]. The hypothesis is not vacuous: that paper’s non-vacuity lemma supplies a nonzero observable satisfying it whenever the state space has two points.

What is uniform and what is not. The *rate* is uniform in L . The constant C_A is not claimed to be, and depends on the observable; the threshold N_0 does not depend on the observable, by the modulus paper’s uniform-threshold theorem, but is not claimed uniform in L either.

6 Scope, and a judge that failed

This is a product measure. At constant source weight the spatial slices decouple, so the Gibbs measure is a product across sites and the statement above is, in substance, the one-dimensional chain repeated. That is why it is reachable. What it establishes is narrower than it may sound: not that the coupled model clusters uniformly, but that the *obstruction to uniformity is the source weight and not the extent*.

The coupled kernel is exactly a 2D Ising transfer matrix. With the ring source weight, $K = D_w^{1/2} T D_w^{1/2}$ where T is the inter-slice product and D_w the intra-slice ring weight — which is the symmetrised transfer matrix of the two-dimensional Ising model, vertical coupling β and horizontal coupling γ . This is an exact algebraic identification, not a numerical claim.

And a pre-registered judge that failed. Before any of the above was written we committed two falsifiable predictions with their thresholds fixed in advance `scripts/judge_spatial_uniform.py`. The first — the decoupled rate is exactly $\tanh \beta$ at every extent — passed to 10^{-16} and authorised this paper. The second — that the *coupled* uniformity boundary is the Onsager curve $\sinh 2\beta \sinh 2\gamma = 1$ — predicted saturation below 1 on the five disordered cells and a rate tending to 1 on the three ordered ones. Seven of eight cells agreed. The cell $(\beta, \gamma) = (0.1, 1.0)$, disordered, gave `specRatio(12) = 0.6266` — nowhere near 1 — but with drift from $L = 10$ of 3.5×10^{-2} , above the pre-registered 10^{-2} , so the cell is *undecided at $L \leq 12$* and the judge does not pass.

We did not raise the extent until the cell passed. The autopsy `scripts/judge_spatial_uniform_autopsy.py` prints the whole sequence instead: its increments decay geometrically

with ratio 0.835, extrapolating to about 0.707, so this is saturation not yet reached and not a crawl to 1. That is a diagnosis, not a pass. **The Onsager identification of the boundary is therefore reported as not established**, and no statement in this paper rests on it.

No claim about the physical target. Reflection positivity is untouched, and nothing here concerns $SU(N)$, the continuum limit, or the Clay problem [6]. The distance to that problem is unchanged.

7 What is verified and not proved

The measured $\text{specRatio}(L)$ for the coupled kernel, extended past the five points the modulus paper printed:

(β, γ)	$L = 1$	$L = 12$	increments
(0.2, 0.2)	0.1974	0.294449	geometric, ratio 0.278
(0.3, 0.4)	0.2913	0.646895	geometric, ratio 0.648
(0.1, 1.0)	0.0997	0.626639	geometric, ratio 0.835
(0.8, 1.2)	0.6640	1.000000	to 1

These are **verified** and **not proved**; no theorem depends on them. They are printed because they are the reason the coupled case is believed hard rather than false, and because the last row is the one the whole lane has been reporting as its limitation.

8 Reproducibility

All artifacts are at source checkpoint `af88cd6abc95c871495f7ac911690e938dc177b3` on branch `main`. Every hyperlink was resolved against that commit, and checked to point at an actual declaration line, before compilation.

Quantity	Value
Full core build	8431 jobs, success
Oracle commands, and answers	2689
distinct declarations	2686
commands listed twice	3
distinct, with axiom dependencies	2663
distinct, axiom-free	23
Declarations in this module	17
Distinct axioms across all outputs	<code>propext</code> , <code>Quot.sound</code> , <code>Classical.choice</code>
Occurrences of <code>sorryAx</code>	0
Project axioms	0
Errors	0

The module is `YangMills/OS/SpatialUniform.lean`; the pre-registered judge and its autopsy are `scripts/judge_spatial_uniform.py` and `scripts/judge_spatial_uniform_autopsy.py`; verdicts are in `docs/VERIFICATION-LEDGER.md`.

References

- [1] L. Eriksson, *Where the Elementary Reconstruction Stops: a Machine-Checked Obstruction at the Coupled $\mathbb{Z}_2$ Slice*, viXra:2607.0075, 2026.
- [2] L. Eriksson, *Strict but Not Uniform: a Machine-Checked Spectral Gap at Every Finite Extent*, viXra, 2026.
- [3] L. Eriksson, *The Modulus: a Machine-Checked Operator Bound on the Fluctuation Sector of the Coupled $\mathbb{Z}_2$ Slice*, viXra, 2026.
- [4] L. Onsager, *Crystal statistics I: a two-dimensional model with an order-disorder transition*, Phys. Rev. **65** (1944), 117–149.
- [5] E. Seneta, *Non-negative Matrices and Markov Chains*, 2nd ed., Springer, 1981.
- [6] A. Jaffe and E. Witten, *Quantum Yang–Mills theory*, Clay Mathematics Institute Millennium Prize problem description, 2000.